

Multi-ancestry finemapping of gene clusters implicated in cardiovascular disease and mental health

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2026-01-31

A genomic region containing a cluster of genes that regulate fatty acid desaturation has recently been found to have a strong association with major depressive disorder¹. However, it is unclear which variant in this region is the cause of the association. This project will use biobank whole genome-sequencing datasets to finemap common variants, catalogue rare variants, and understand shared genetic risk between cardiovascular disease and common psychiatric disorders.

The FADS gene cluster

The fatty acid desaturase gene family (FADS1, FADS2, FADS3) is co-located in a gene-dense region on chromosome 11 (11q12-q13.1). FADS1 and FADS2 regulate the conversion of polyunsaturated fatty acids (omega-6 and -3) into long-chain forms, which have roles in a wide range of cellular processes and are implicated in cardiovascular, metabolic, and immunological health².

The role of FADS as the genes underlying the association with major depression is bolstered by a genetic causal analysis (Mendelian randomisation) of circulating metabolites³, which

1. [Adams, M.J. et al. *Cell* **188**, 640–652 \(2025\).](#)
2. [Koletzko, B. et al. *Annual Review of Nutrition* **39**, 21–44 \(2019\).](#)
3. [Davyson, E. et al. *Biological Psychiatry* **94**, 630–639 \(2023\).](#)

showed that omega-3 levels and omega-6 to omega-3 ratios had a causal effect on depression. While this makes FADS1 and FADS2 the most plausible biological candidates, expression levels of other genes in this region (MYRF and TMEM258) also have possible links to depression. Further work is therefore required to pinpoint the causal variant driving the association.

Genetic associations

Prior studies examining specific FADS genetic variants and depression^{4,5} had small sample sizes ($N \approx 200$ –1000). Large genome-wide association studies (GWASs) have identified a strong association signal in this region, but the most significant variant has been located in other nearby genes like DAGLA^{6,7} and TMEM258^{8,9}. However, the most recent and largest multi-ancestry genetic study of depression found the top signal to be a single nucleotide intron variant in FADS2 (Figure 1).

Fine-mapping

Because genetic variants in a region are in close physical linkage, association studies find a correlated statistical signal across variants that makes it difficult to determine which variant is the cause of the association.

Acknowledgments

This work was supported by a BHF REA4 Pump Priming Award (RE/24/130012).

4. [Cribb, L. et al. *Nutritional Neuroscience* **21**, 589–601 \(2018\).](#)
5. [Sublette, M.E. et al. *Psychiatric Genetics* **26**, \(2016\).](#)
6. [Howard, D.M. et al. *Nature Neuroscience* **22**, 343–352 \(2019\).](#)
7. [Meng, X. et al. *Nature Genetics* \(2024\).doi:10.1038/s41588-023-01596-4](#)
8. [Levey, D.F. et al. *Nature Neuroscience* **24**, 954–963 \(2021\).](#)
9. [Als, T.D. et al. *Nature Medicine* **29**, 1832–1844 \(2023\).](#)

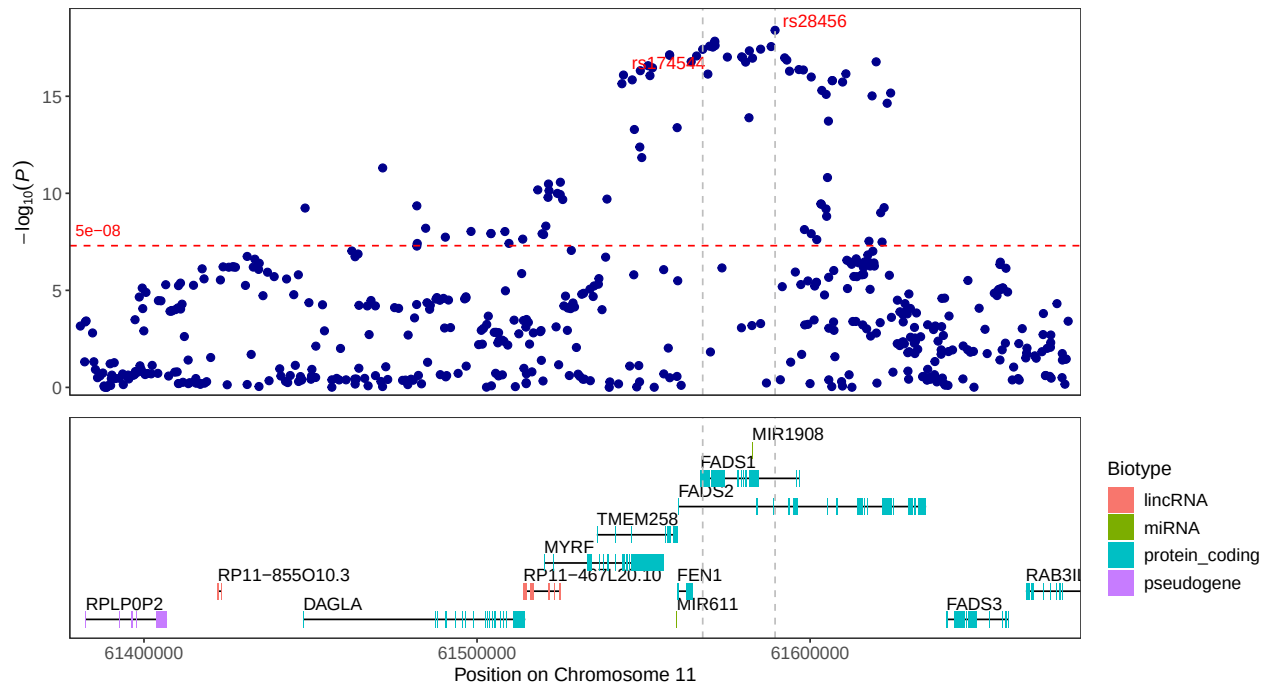


Figure 1: Region plot of genome-wide association findings for the FADS gene cluster. Top: Tested variants with transformed association p-value. Bottom: Gene positions in the region.